Sina Aghaee Dabaghan Fard

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Education

Ph.D., Industrial & Systems Engineering (Data Science Track), **Texas A&M University**, College Station, TX, USA. 2023 – Present

M.S., Industrial & Systems Engineering (Engineering Management Track), **Sharif University of Technology**, Tehran, Iran. 2023

B.S., Industrial & Systems Engineering, **Iran University of Science and Technology**, Tehran, Iran. 2020

Current Research Areas

- Bayesian and physics-informed machine learning
- Quality engineering and prognostics for complex engineering systems
- Data-driven analytics for IoT-enabled smart manufacturing systems

Publications

Published/Accepted

1. **S. Aghaee Dabaghan Fard**, M. Kim, A. Deep, J. Lee. "*Bayesian Joint Model of Multi-Sensor and Failure Event Data for Multi-Mode Failure Prediction.*" *Technometrics*, 2026. DOI: 10.1080/00401706.2026.2653564.

   ** This paper was a finalist for the QCRE Best Student Paper Competition, 2025 IISE Annual Conference & Expo.*

2. T. Srikitrungruang, **S. Aghaee Dabaghan Fard**, M. Lemon, J. Lee, Y. Zhou. "*Robust Physics-Informed Neural Network Approach for Estimating Heterogeneous Elastic Properties from Noisy Displacement Data.*" *Advanced Science*, 2026. DOI: 10.1002/advs.202508445.

   ** Advanced Science is a premier interdisciplinary journal with high impact (IF > 14).*

Under Review/Revision

1. **S. Aghaee Dabaghan Fard**, M. Maros, J. Lee. "*Efficient Bayesian Deep Ensembles via Analytic Predictive Inference.*" arXiv: 2607.06776.

In Preparation

1. **S. Aghaee Dabaghan Fard**, J. Lee. "*BEIFSurv: Bayesian Ensemble-Induced Finite Function-Space Survival Model for Multi-Sensor Data and Multi-Mode Failure Prediction.*"

2. **S. Aghaee Dabaghan Fard**, J. Lee. "*BLL-PINN: Bayesian Last-Layer Physics-Informed Neural Networks.*"

Teaching Experience

Teaching Assistant, **Texas A&M University**, College Station, TX, USA Aug 2023 – Dec 2024
– ISEN 350: Quality Engineering — Fall 2023 (45 students), Fall 2024 (75 students); (Undergraduate course).
  * Instructed and supervised laboratory sessions
– ISEN 614: Advanced Quality Control — Spring 2024 (Graduate course).
– DAEN 210: Uncertainty Modeling — Spring 2024, Spring 2025 (Undergraduate course).

Teaching Assistant, **Sharif University of Technology**, Tehran, Iran Sep 2021 – Jul 2023
– SSE: Software Systems Engineering — Spring 2023 (Graduate course).
– MIS: Management Information Systems — 2021–2022 (Undergraduate course).
– CIS: Computer and Information Systems — 2021–2022 (Undergraduate course).

Academic & Professional Experience

Graduate Research Assistant, **Texas A&M University**, College Station, TX Aug 2023 – Present
– Conduct advanced statistical modeling and data analysis using Bayesian inference, Gaussian processes, and machine learning for research in complex engineering systems.
– Establish and validate predictive models, perform simulations, and derive theoretical results to support methodological rigor.
– Prepare technical reports, academic papers, and conference presentations.

- **Industry Collaboration: Crider Foods** Feb 2025 – Mar 2026

  – As part of the Texas A&M Engineering Experiment Station (TEES) team, collaborated with Crider Foods to develop smart monitoring solutions for can seaming operations within an Industry 4.0 framework.
  – Assisted with the installation and calibration of IoT sensors for reliable real-time multi-sensor data acquisition.
  – Designed and implemented automated data labeling and monitoring pipelines using statistical quality control methods for high-frequency sensor data streams.
  – Developed a real-time machine breakdown prediction system using transformer, CNN, and deep neural network architectures to predict downtime events caused by multiple failure modes, including lid-missing, dented cans, and seamer jams.

Business Analyst, **Danesh Co.**, Tehran, Iran Feb 2021 – Jun 2023
Contributed to software projects in the finance industry: Order Management System (OMS), financial data processing platform, and trading algorithm development.

Business Analyst (Intern), **PodroCo**, Tehran, Iran Nov 2020 – Dec 2020
– Conducted market research and data analysis for an international project in the Middle East market.
– Identified key startups and industry trends through web scraping and user research.

Research Assistant, **Iran University of Science and Technology**, Tehran, Iran Dec 2019 – Mar 2020
Conducted process simulation and value stream mapping to optimize patient flow and reduce bottlenecks in the Shohadaye Tajrish Hospital Emergency Department.

Presentations & Posters

Conference & Workshop Presentations

– "Bayesian Joint Model of Multi-Sensor and Failure Event Data for Multi-Mode Failure Prediction," CRSE Workshop, Wharton School, University of Pennsylvania, Philadelphia, PA, USA, August 2026 (Selected Participant)

– "BLL-PINN: Bayesian Last-Layer Physics-Informed Neural Networks for Uncertainty Quantification," INFORMS Annual Meeting, Atlanta, GA, USA, October 2025

– "Joint Model for Multi-Type Failure Event Prediction from Multi-Sensor Time Series and Survival Data," IISE Annual Conference & Expo, Atlanta, GA, USA, May 2025 (*Finalist, QCRE Best Student Paper Competition*)

– "Joint Model for Multi-Type Failure Event Prediction from Multi-Sensor Time Series and Survival Data," INFORMS Annual Meeting, Seattle, WA, USA, October 2024

Posters

– "Bayesian Joint Model of Multi-Sensor and Failure Event Data for Multi-Mode Failure Prediction," CRSE Workshop, Wharton School, University of Pennsylvania, Philadelphia, PA, USA, August 2026 (Selected Participant)

– "Joint Model for Multi-Type Failure Event Prediction from Multi-Sensor Time Series and Survival Data," TAMU ISEN Poster Competition, College Station, TX, USA, March 2025

– "Joint Model for Multi-Type Failure Event Prediction from Multi-Sensor Time Series and Survival Data," INFORMS Annual Meeting, Seattle, WA, USA, October 2024

– "Joint Model for Multi-Type Failure Event Prediction from Multi-Sensor Time Series and Survival Data," TAMU Zorich's Reliability Workshop, College Station, TX, USA, September 2024

Awards & Achievements

– **CRSE Workshop, Selected Oral Presenter**, Wharton School, University of Pennsylvania (2026); travel award recipient
– **Industrial & Systems Engineering Travel Award**, Texas A&M University (2024, 2025)
– **Best Student Paper Competition, Finalist**, IISE Annual Conference & Expo (2025)
– **IISE Abstract Contest, 2nd Place Winner**, ISEN Writing and Student Success Center, Texas A&M University (2024)

Paper Review Referee

– Journal of Intelligent Manufacturing

Professional Memberships

– Institute for Operations Research and the Management Sciences (INFORMS).
– Institute of Industrial and Systems Engineers (IISE).
– American Statistical Association (ASA).

Skills

Programming Languages:	Python (PyTorch, Pandas, NumPy, and related libraries), R, SQL, MATLAB
Software & Tools:	Minitab, Vensim, Bizagi, MS Office, Power BI, Visual Paradigm, LaTeX, Git, GitHub, GitLab, Markdown
Analytical & Modeling Skills:	Data analytics, Machine & deep learning, Bayesian modeling & inference, Physics-Informed Neural Networks (PINNs), Survival analysis, Gaussian processes, Uncertainty quantification, Statistical quality control

Last updated: September 1, 2026